

MQAScore: FINE-GRAINED TEXT-MUSIC ALIGNMENT EVALUATION VIA LARGE AUDIO LANGUAGE MODEL

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ABSTRACT

Text-to-music alignment is commonly evaluated by the global similarity between text and audio embeddings, which collapses all described attributes into a single score and obscures which ones are missing. We instead introduce MQAScore, which verifies each atomic musical concept along five clip-level dimensions (genre, instrument, mood/theme, vocal, and usage) by the probability that a large audio language model (LALM) answers affirmatively to a binary question. To assess evaluator reliability, we construct verification labels from presence/absence annotations and LLM-guided concept substitutions, and propose the Sensitivity-Invariance Ratio, which measures how selectively an evaluator responds to meaning-changing substitutions. Experiments show that concept-level scoring improves both question-answering and similarity-based evaluators and MQAScore can be a more reliable and interpretable fine-grained complement to global similarity for evaluating text-to-music alignment.

Index Terms— Text-to-music evaluation, text–music alignment, large audio language models

1. INTRODUCTION

The field of music generation has evolved rapidly, with state-of-the-art text-to-music (TTM) systems now capable of generating high-fidelity and vocalized music. Alongside these advancements, text remains the primary conditioning modality, with prompts becoming more intricate to improve controllability. To evaluate text-audio alignment, existing studies predominantly rely on similarity-based metrics [1, 2] computing a global similarity score between text and audio embeddings. However, as shown in Fig 1, a global similarity score summarizes all attributes in the prompt and audio into single value, obscuring fine-grained semantic information and making the evaluation difficult to interpret. As text-to-music generation moves toward detailed, multi-attribute prompts, alignment evaluation should move beyond global similarity and focus on evaluating individual semantic concepts.

Moving toward such fine-grained concept-level evaluation raises two challenges. The first is to define the evaluation space, as musical captions span attributes ranging from descriptive (e.g., genre, instrument) to musicological (e.g., key,

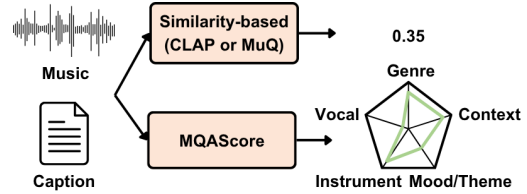


Fig. 1. Global similarity (e.g. CLAP, MuQ) yields a single score; MQAScore verifies each musical dimension separately.

chord, tempo). Our taxonomy follows CAL500 [3], whose tags were collected from non-expert listeners and span six dimensions: emotion, genre, instrument, vocal, usage, and acoustic qualities. We retain the first five, renaming emotion as mood/theme, i.e., the intended situational setting. These dimensions remain standard in music tagging [4, 5], and their concepts are categorical, so the presence or absence of each concept in a clip is well defined, although detection difficulty varies across dimensions [6]. We exclude acoustic qualities, which reflect audio fidelity rather than semantic alignment, as well as the expert-level musicological concept of Pandora’s Music Genome Project¹ (e.g., harmony, rhythm), since both remain difficult for LALMs to perceive [7, 8].

The second challenge is cost: overall human ratings of text-to-music systems already require thousands of annotators and comparisons [9], and concept-level annotation multiplies this cost by the number of attributes per prompt. LALMs offer a scalable alternative given their rapidly improving music understanding [10–12], yet remain unreliable at open-ended musical reasoning [8, 13]. Following recent text-to-audio alignment metrics [14–16], we therefore restrict the LALM to deciding whether a single concept is supported by the clip. MQAScore adopts the scoring mechanism of AQAScore [14], the probability of an affirmative answer, but applies it to concept-level rather than to the full prompt. Since captions reveal which concepts are present but not which are absent, we obtain negatives from present/absent annotations for genre and instrument. For mood/theme, vocal, and usage, by replacing each concept with its most distant alternative under an LLM’s text prior, using the closest alternative as a meaning-preserving control. We then assess MQAScore

¹<https://www.pandora.com/corporate/mgp.shtml>

by how much more it responds to contradicting substitutes than to meaning-preserving ones, and by its agreement with experts in pairwise comparisons.

In summary, our contributions are threefold: 1) We introduce a taxonomy of five clip-level dimensions grounded in CAL500, together with MQAScore, which verifies each atomic musical concept separately with an LALM. 2) We construct verification labels from presence/absence annotations and LLM-guided concept substitutions, and propose the Sensitivity-Invariance Ratio ($R_{S/I}$) to measure how selectively an evaluator responds to meaning-changing substitutions. 3) We show that concept-level decomposition improves both similarity-based and question-answering-based scores, and that LALM-based scoring further outperforms similarity-based scoring in both discrimination and human agreement.

2. RELATED WORK

Despite the rapid advancement of music generation models, automatic evaluation of text-music alignment remains challenging. Conventional text-audio and text-music alignment methods largely rely on global similarity in joint cross-modal embedding spaces, such as CLAP-based scores, and MuQ-MuLan [1, 2, 17]. Studies have highlighted the limitations of such global representations in capturing detailed musical attributes [18], while finer-grained evaluation methods have emerged to assess individual acoustic events, musical attributes, or compositional constraints [19, 20]. Evaluation research has also emphasized stronger agreement with human judgments through subjective alignment ratings, expert assessments, mean opinion scores, or preference annotations [9, 20–22]. Meanwhile, music question-answering datasets and benchmarks have been developed to probe LALM capabilities in music understanding, instruction following, and reasoning [6, 8, 23, 24], while recent general and music-specialized LALMs, including Audio Flamingo Next, Music Flamingo, and Qwen3-Omni, demonstrate increasingly strong capabilities in these areas [10–12]. Building on these capabilities, emerging approaches have begun repurposing LALMs themselves as semantic evaluators, reformulating audio–text alignment as question answering, rubric-based verification, or concept-level judging [14–16, 25].

3. METHOD

3.1. Taxonomy Construction

Following CAL500 [3], we adopt five dimensions as

$$\mathcal{D} = \{\text{Genre, Instrument, Mood/Theme, Vocal, Usage}\},$$

renaming emotion as mood/theme and excluding acoustic qualities. For each caption c in MusicCaps [26], an LLM extracts clip-level concepts $\mathcal{E}(c) = \{(d_i, t_i)\}_{i=1}^K$, assigning each atomic concept t_i to a dimension $d_i \in \mathcal{D}$ and discarding

subjective modifiers (e.g., “groovy bass” yields “bass”). The vocabulary $\mathcal{V}_d$ collects the concepts extracted for dimension d (e.g., $\text{bass} \in \mathcal{V}_{\text{Instrument}}$), after merging synonyms and manual review.

Verification Label Construction. Verifying a concept requires both positives, i.e., concepts supported by the clip, and negatives, i.e., concepts not supported by it. Given a clip a and a concept $t \in \mathcal{V}_d$ supported by a , we construct negatives $t^- \in \mathcal{V}_d$ in a dimension-dependent manner, and in all dimensions a reliable evaluator should score t above t^- . For genre and instrument, we pair t with each concept annotated as absent from a [27, 28], forming triplets (a, t, t^-) . For mood/theme, vocal, and usage, such annotations are unavailable, and concepts are not mutually exclusive (e.g., a clip can be both calm and melancholic), so an arbitrary substitute may still be supported. We therefore replace t with its semantically most distant concept in $\mathcal{V}_d$, as judged by an LLM, to obtain t^- . Since this substitution alters the text itself, a lower score for t^- may reflect the edit rather than the audio; we thus also take the closest concept as a concept-preserving substitute $t^+ \in \mathcal{V}_d$, forming quadruplets (a, t, t^+, t^-) . For these dimensions, a reliable evaluator should additionally respond little to t^+ ; since t^+ and t^- undergo the same substitution, a stronger response to t^- than to t^+ indicates that the score change stems from the audio rather than the edit.

3.2. Concept Verification Scoring

Following AQAScore [14], we cast evaluation as audio-grounded binary question answering, but query each concept separately rather than the full prompt. For a concept $t \in \mathcal{V}_d$, a dimension-specific template ϕ_d produces a question $q = \phi_d(t)$ (e.g., Does this music belong to the genre $\{t\}$? Answer Yes or No.), and an LALM p_θ scores a clip a by the probability of “Yes” normalized against “No”:

$$S(a, t) = \frac{p_\theta(\text{“Yes”} \mid a, q)}{p_\theta(\text{“Yes”} \mid a, q) + p_\theta(\text{“No”} \mid a, q)},$$

which we refer to as MQAScore. Given the concepts $\mathcal{E}(c) = \{(d_i, t_i)\}_{i=1}^K$ extracted from a caption c , the alignment $S_d(a, c)$ along dimension $d \in \mathcal{D}$ is the mean of $S(a, t_i)$ over $\mathcal{I}_d = \{i \mid d_i = d\}$, yielding a per-dimension score vector rather than a single global score; dimensions with $\mathcal{I}_d = \emptyset$ are reported as unavailable rather than assigned zero.

4. EXPERIMENTS

4.1. Dataset

Genre, Instrument. GTZAN [27] contains 1,000 30-second clips evenly distributed over 10 genres. As GTZAN is single-labeled, we take each clip’s genre as t and treat the remaining nine genres as negatives, yielding nine triplets (a, t, t^-) per

clip. OpenMIC-2018 [28] provides 20,000 10-second clips with partial annotations over 20 instruments, including explicit absence. Following the official release, we label an annotated instrument present if its relevance exceeds 0.5 and absent otherwise. Pairing each present instrument with every absent one in the same clip, excluding voice, yields 15,241 triplets over 7,037 clips.

Mood/Theme, Vocal, and Usage. We apply the concept extraction in Sec. 3.1 on MusicCaps [26] and construct t^+ and t^- for each extracted concept yielding 3967, 737, and 112 quadruplets (a, t, t^+, t^-) for mood/theme, vocal, and usage, respectively. The LLM we used for all the textual processing is Qwen3-4B-Instruct-2507.

Expert Pairwise Annotations. To test whether MQAScore ranks pairs as experts do, we sample clips from MusicCaps and have 11 annotators with musical training label pairs following written guidelines. For genre and instrument, given a concept t and two clips (a_1, a_2) , annotators label which clip exhibits t . For mood/theme, vocal, and usage, given a clip a and a pair (t, t^-) , annotators label which concept applies to a . The label set also includes “both” and “neither”; we retain only pairs with an unambiguous label, yielding 147 and 119 pairs for the two types, respectively, with an inter-annotator agreement of Krippendorff’s $\alpha = 0.71$.

4.2. Evaluators

We evaluate two scoring paradigms at two input granularities. At the *caption-level*, an evaluator scores the full caption c , and a substitute is evaluated by replacing t with t^+ or t^- within c . At the *concept-level* level, each extracted concept is scored in isolation as $S(a, t)$. Caption-level scoring applies only to mood/theme, vocal, and usage, as GTZAN and OpenMIC-2018 provide tag annotations rather than captions.

Similarity-based. We use two contrastive text–audio models, LAION-CLAP [1] (music_audioset_epoch_15_esc_90.14.pt) and MuQ-MuLan [2] (OpenMuQ/MuQ-MuLan-large), and take the cosine similarity between the text and audio embeddings as the score. At the concept level, each concept is embedded with the template “This is a music piece about {t}.”

QA-based. We use three LALMs: Qwen3-Omni [10] (Qwen3-Omni-30B-A3B-Instruct), Audio Flamingo Next [12], and Music Flamingo [11]. At the caption level, we apply AQAScore [14], which asks whether the clip matches the full caption; at the concept level, we apply MQAScore. Scores are computed from the next-token probabilities of “Yes” and “No” in a single forward pass, without sampling.

4.3. Evaluation Metrics

AUC. For each dimension d , we pool the scores of all positives $S(a, t)$ and negatives $S(a, t^-)$ across clips and report the unpaired ROC-AUC (Mann–Whitney U), with ties counted as

0.5. As positives and negatives are not paired within a clip, it also penalizes evaluators whose scores drift across clips.

Win Rate (WR). For genre and instrument, we further measure within-clip discrimination. For each clip, we compute the fraction of (present, absent) concept pairs in which the present instrument is scored higher, and average this fraction over clips with at least one present and one absent annotation.

Sensitivity-Invariance Ratio ($R_{S/I}$). For mood/theme, vocal, and usage, we measure how much scores change under substitution by the normalized mean absolute difference (z -MAD) over quadruplets (a, t, t^+, t^-) :

$$z\text{-MAD}^k = \mathbb{E}_{(a, t, t^k)} \left[\left| \tilde{S}(a, t) - \tilde{S}(a, t^k) \right| \right], \quad k \in \{-, +\},$$

where $\tilde{S}(a, t) = (S(a, t) - \mu_S) / \sigma_S$ is the z -normalized score, with μ_S and σ_S the mean and standard deviation of each evaluator’s scores on the original concepts. We then define $R_{S/I} = z\text{-MAD}^- / z\text{-MAD}^+$, which quantifies how much further an evaluator’s score moves when the concept is contradicted than when it is replaced by a close alternative; $R_{S/I} \gg 1$ indicates that its judgments follow the audio rather than the wording, whereas $R_{S/I} \approx 1$ means it cannot tell a contradiction from a near-synonym.

Agreement with Expert Labels. The expert labels serve as ground truth for each pair. For an evaluator f , we derive its prediction for a pair from its scores as the option with the higher score, namely $\arg \max_{a' \in \{a_1, a_2\}} S(a', t)$ for genre and instrument and $\arg \max_{t' \in \{t, t^-\}} S(a, t')$ for mood/theme, vocal, and usage, with ties counted as disagreement. The human-evaluator agreement is the fraction of expert labels matching the evaluator’s prediction. where i indexes the N individual expert labels, $y_{\text{human}}^{(i)}$ is the i -th label, and $\hat{y}_f^{(i)}$ is the prediction of f .

5. RESULTS

Concept Verification. Table 1 shows that concept-level scoring improves both paradigms over caption-level scoring, as isolating each concept prevents its contribution from being diluted by the other concepts in a caption. At the concept level, LALMs further outperform similarity-based evaluators on mood/theme, vocal, and usage in both AUC and $R_{S/I}$. Similarity-based evaluators respond almost as strongly to close alternatives as to contradictions, with MuQ even reacting more to close alternatives on vocal, suggesting that their scores track the wording rather than the audio. On genre and instrument, similarity-based evaluators remain competitive, likely because these closed vocabularies are well covered by contrastive pretraining. Caption-level AQAScore shows a high $R_{S/I}$ despite its lower AUC, as replacing a single concept in a multi-concept caption barely moves the score under a close alternative, inflating the ratio. Usage is the most challenging dimension for all evaluators, as situational context is inferred from, rather than directly heard in, the audio. Finally,

Table 1. Concept verification results: AUC with WR for genre and instrument, and with $R_{S/I}$ for mood/theme, vocal, and usage. Caption-level scores are unavailable for tag-only genre and instrument data.

Input	Method	Model	Genre		Instrument		Mood/Theme		Vocal		Usage	
			AUC	WR	AUC	WR	AUC	$R_{S/I}$	AUC	$R_{S/I}$	AUC	$R_{S/I}$
Caption	AQAScore	Qwen3-Omni	–	–	–	–	0.776	31.3	0.870	31.0	0.718	36.9
		AF-Next	–	–	–	–	0.739	20.6	0.727	9.4	0.685	6.9
		Music Flamingo	–	–	–	–	0.684	10.0	0.711	6.8	0.610	5.1
	Similarity	CLAP	–	–	–	–	0.535	1.94	0.629	2.63	0.559	1.47
		MuQ	–	–	–	–	0.532	2.34	0.500	0.38	0.528	1.55
	Concept	MQAScore	Qwen3-Omni	0.983	0.978	0.915	0.887	0.943	7.38	0.966	21.90	0.842
AF-Next			0.980	0.972	0.893	0.849	0.900	5.12	0.881	7.03	0.744	4.43
Music Flamingo			0.982	0.977	0.855	0.797	0.885	3.22	0.886	3.28	0.713	3.03
Similarity		CLAP	0.912	0.930	0.867	0.849	0.669	1.91	0.759	2.06	0.711	1.39
		MuQ	0.959	0.946	0.860	0.812	0.663	1.73	0.511	0.48	0.654	1.53

Table 2. Agreement (%) with expert labels on MusicCaps pairs. H: human–human agreement; Q3-O: Qwen3-Omni; AF-N: Audio Flamingo Next; MF: Music Flamingo. All evaluators use concept-level scoring.

	H	Q3-O	AF-N	MF	CLAP	MuQ
Genre	92.5	85.0	86.7	75.0	71.7	71.7
Instrument	82.3	79.7	72.9	89.8	74.6	62.7
Mood/Theme	84.6	86.4	88.6	88.6	54.5	43.2
Vocal	95.2	92.5	92.5	85.0	80.0	60.0
Usage	82.5	85.4	78.0	68.3	63.4	61.0

evaluators with similar AUC can differ markedly in $R_{S/I}$, indicating that discrimination accuracy alone does not capture how selectively an evaluator responds to meaning-changing substitutions.

Agreement with Expert Labels. Table 2 compares evaluators with expert labels, using human-human agreement as a reference. LALM-based evaluators reach agreement comparable to that among humans across dimensions, whereas similarity-based evaluators fall far behind, most notably on mood/theme, where MuQ drops below chance. The gap is smaller on genre and instrument, likely because these closed vocabularies are well covered by contrastive pretraining. Abstract concepts such as mood and usage, in contrast, benefit from the LLM backbone of LALMs, which interprets concept semantics beyond what a single embedding match captures, and from explicitly querying the clip for each concept. Since all evaluators here score concepts individually, this advantage stems from question answering rather than granularity alone. Although rankings among LALMs vary, the gap between LALMs and similarity-based evaluators is consistent.

6. CONCLUSION

We presented MQAScore, which evaluates text-to-music alignment by verifying atomic musical concepts along five dimensions with LALMs, and assessed it with dimension-specific verification labels and expert annotations. Concept-level scoring improves both question-answering and similarity-based evaluators over caption-level scoring, and at the concept level, LALMs outperform similarity-based evaluators and reach human–human agreement on mood/theme, vocal, and usage. Our Sensitivity–Invariance Ratio further shows that evaluators with similar AUC can differ markedly in how selectively they respond to meaning-changing substitutions, and usage remains the most challenging dimension. Our study has several limitations. Negatives for mood/theme, vocal, and usage are selected by an LLM based on textual distance, which cannot guarantee that every substitute differs significantly and consistently from the reference concept in the audio. GTZAN negatives rely on a single-label assumption that may penalize overlapping genres. Our scope is limited to clip-level descriptive concepts, and our expert set is small and drawn from real rather than generated music. Future work will extend the taxonomy into a graph linking concepts with their modifiers and relations (e.g., groovy → bass), scale verification to musicological attributes such as those in Pandora’s Music Genome Project, and apply concept verification to generated music and as a reward signal for text-to-music generation.

7. COMPLIANCE WITH ETHICAL STANDARDS

This research utilizes data made available in open-access datasets (MusicCaps, GTZAN, and OpenMIC-2018). The expert pairwise annotation involved standard perceptual labeling of audio concepts. As no personally identifiable information (PII) or sensitive data was collected from the

annotators, formal ethical approval was not required by the institutional review board.

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